

Software

Syllabus

4.1 Types of Software & Interrupts

- 1 Describe the difference between system software and application software and provide examples of each
 - System software provides the services that the computer requires, including operating system and utility software
 - Application software provides the services that the user requires
- 2 Describe the role and basic functions of an operating system
 - Including:
 - Managing files
 - Handling interrupts
 - Providing an interface
 - Managing peripherals and drivers
 - Managing memory
 - Managing multitasking
 - Providing a Platform for running applications
 - Providing system security
 - Managing user accounts
 - Applications are run on the operating system
 - The operating system is run on the firmware
 - The bootloader (firmware) is run on the hardware
- 3 Understand how hardware, firmware and an operating system are required to run applications software
- 4 Describe the role and operation of interrupts
 - Including:
 - How an interrupt is generated
 - How it is handled using an interrupt service routine
 - What happens as a result of the interrupts
 - Software interrupts include division by zero and two processes trying to access the same memory location
 - Hardware interrupts include pressing a key on the keyboard and moving the mouse



4.2 Types of Programming Language, Translators and Integrated Development Environments (IDEs)

- 1 Explain what is meant by a high-level language and a low-level language, including the advantages and disadvantages of each
 - Advantages and disadvantages include:
 - Ease of reading and writing code, e.g., low-level is hard to read
 - Ease of debugging code
 - Machine independence
 - Direct manipulation of hardware
- 2 Understand that assembly language is a form of low-level language that uses mnemonics, and that an assembler is needed to translate an assembly language program into machine code
- 3 Describe the operation of a compiler and an interpreter, including how high-level language is translated by each and how errors are reported
 - A compiler translates the whole code at once before executing it, producing an executable file
 - An interpreter translates and executes the code line-by-line
 - A compiler provides an error report for the whole code if errors are detected
 - An interpreter stops execution when an error is found
- 4 Explain the advantages and disadvantages of a compiler and an interpreter
 - To include an understanding that an interpreter is mostly used when developing a program and a compiler is used to translate the final program
- 5 Explain the role of an IDE in writing program code and the common functions IDEs provide
 - Including:
 - Code editors
 - Run-time environment
 - Translators
 - Error diagnostics
 - Auto-completion
 - Auto-correction
 - Pretty print

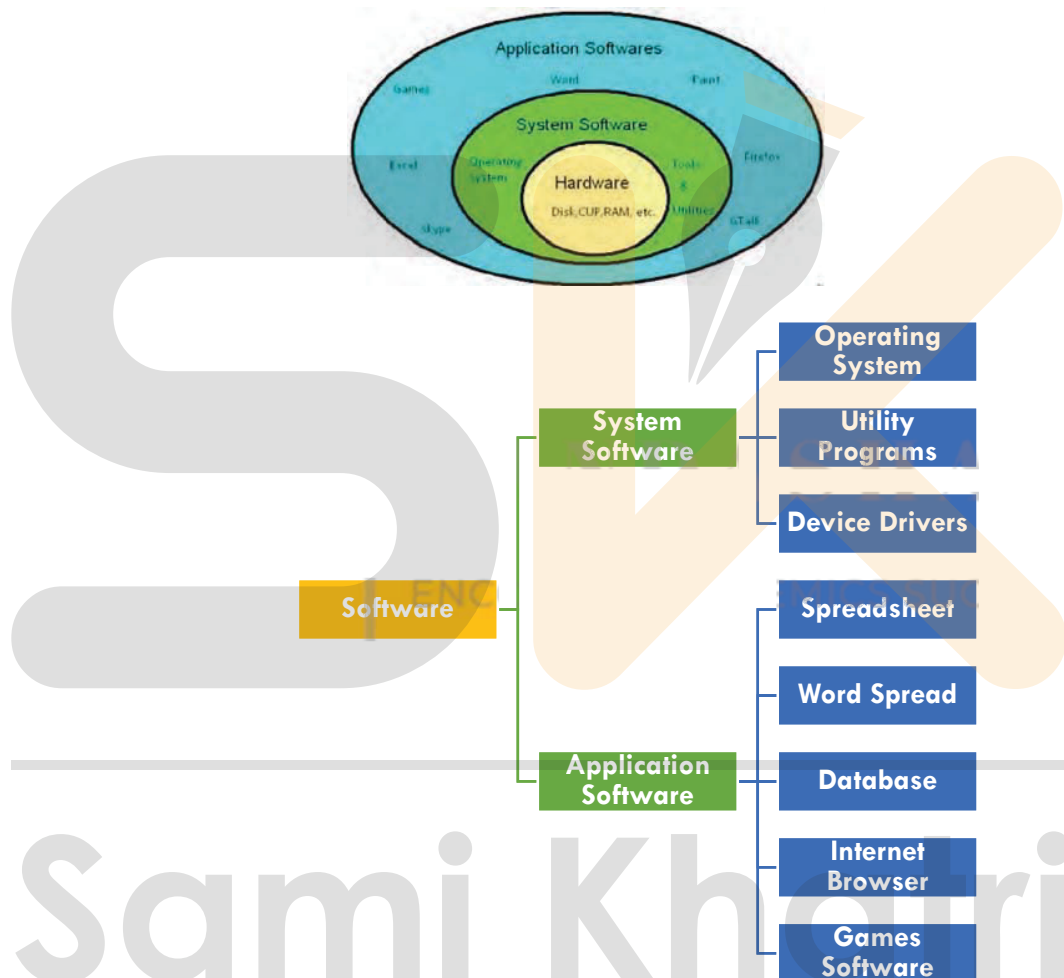


4.1 Types of Software & Interrupts

Software

Set of Instructions (Computer Programs), procedures and documents associated with computer programs and their use & purpose.

Software is the intermediary between hardware and user.



System Software

System software is software designed to provide a platform for other software.

Types of System Software

- Operating System
- Utility Software
- Device Driver



+92-335-8467840



cswithsk@gmail.com

Features

- Set of programs to control and manage the operation of computer hardware
- Platform on which other software can run
- Required to allow hardware and software to run without problems
- Human Computer Interface (HCI)
- Controls the allocation and usage of hardware resources.

Application Software

Computer program designed to carry out a specific task other than one relating to the operation of the computer itself, typically to be used by end-users.

Features

- Perform various applications (Apps) on a computer
- User Friendly
- May be a single program (for example, Paint) or a suite of programs (for example, Microsoft Office Suite)
- User can execute the software as and when they require

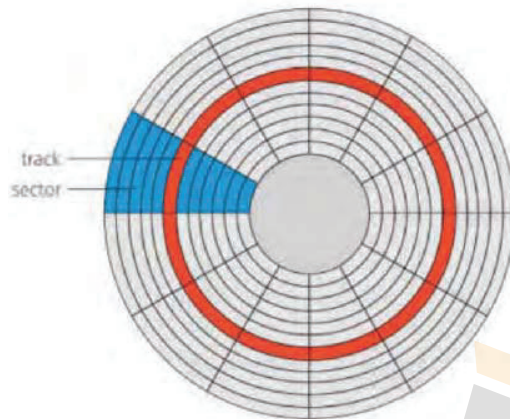
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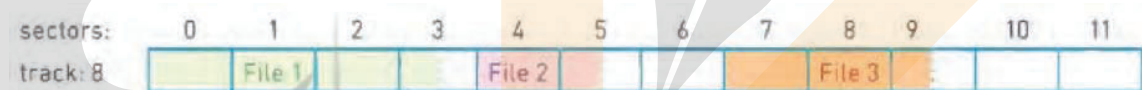
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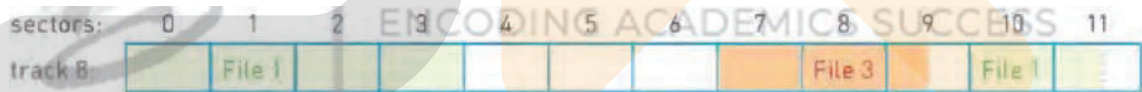
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In this example we have three files (1, 2 and 3) stored on track 8 of a disk surface covering all of 12 sectors:



Now file 2 is deleted by user and file 1 has added to it; however, the file 2 sectors which become vacant are not filled up straight away by new file 1 data since this would require 'too much effort' for the HDD resources; we now get the following (file 1 is now stored in the sectors 0, 1, 2, 3, 10 and 11):



File 1 has now been extended to write data in sectors 10 and 11; now suppose File 3 is extended with the equivalent 3 blocks of data; this now requires filling up sector 9 and then finding some empty sectors to write the remainder of the data – suppose the next free sectors are on track 11:



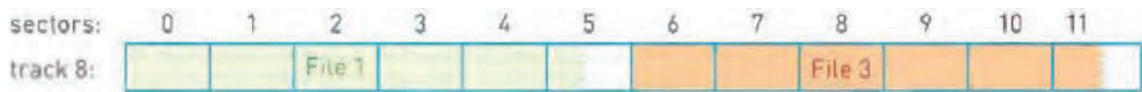
If this continues, the files just become more and more scattered throughout the disk surfaces. It is possible for sectors 4, 5 and 6 (on track 8) to eventually become used if the disk starts to fill up and it has to use up whatever space is available. A disk defragmenter will rearrange the blocks of data to store files in contiguous sectors wherever possible. After defragmentation track 8 would now become:



+92-335-8467840



cswithsk@gmail.com



This obviously allows for much faster data access and retrieval since the HDD will now require fewer read-write head movements to access and read files 1 and 3. Track 11 would be empty after the defragmentation process.

Backup Software

While it is sensible to take manual back-ups using, for example, a memory stick or portable HDD, it is also a good practice to use the operating system back-up utility. This utility will:

- ➔ Allow a schedule for backing up files to be made.
- ➔ Only carry out a backup procedure if there have been any changes made to a file.

For total security there would be three versions of a file:

- 1) The current (working) version stored on the internal HDD or SSD.
- 2) A locally backed up copy of the file (stored on a portable SSD, for example)
- 3) A remote back up version stored well away for the computer (for example, using cloud storage)

The Microsoft Windows environment offers the following facilities using the back-up utility:

- ➔ Restore data, files or the computer from the back-up (useful if there has been a problem and files have been lost and need to be recovered)
- ➔ Create a restore point (this is basically a kind of 'time machine' where your computer can be restored to its state at this earlier point in time; this can be very useful if a very important file has been deleted and can't be recovered by any of the other utilities)
- ➔ Options of where to save back-up files; this can be set up from the utility to ensure files are automatically backed up to a chosen device.

Security Software

- ➔ Security software is an over-arching utility that: for more information on encryption and decryption.
- ➔ Manages access control and user accounts (using user IDs and passwords)
- ➔ Links into other utility software, such as virus checkers and spyware checkers
- ➔ Protects network interfaces (for example, through the use of firewalls)
- ➔ uses encryption and decryption to ensure any intercepted data is meaningless without a decryption key.
- ➔ Oversees the updating of software (does the update request come from a legitimate source, for example).



+92-335-8467840



cswithsk@gmail.com

Screensavers

Screensavers are programs that supply moving and still images on the monitor screen after a period of inactivity by the computer. They were originally developed to protect older CRT (cathode ray tube) monitors which would suffer from 'phosphor burn' if the same screen image remained for any length of time.

Some screensavers are often used to activate useful background tasks that can only go on when the computer is in an 'idle' state. For example:

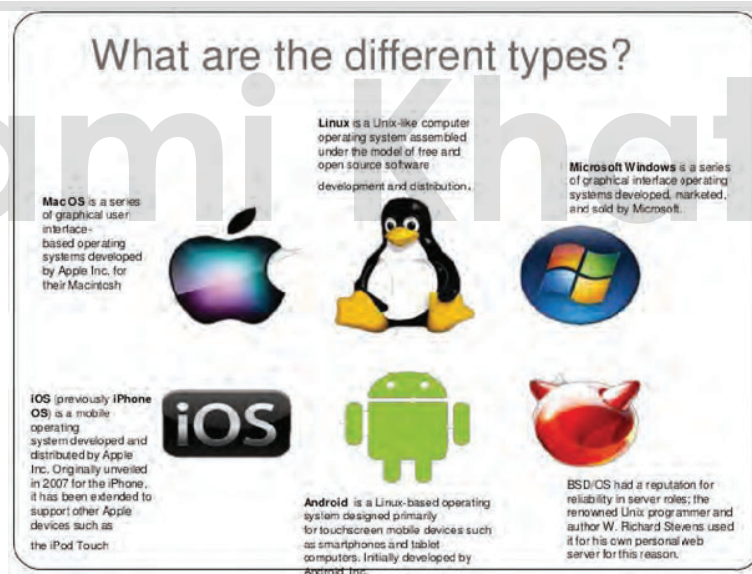
- Scans for virus
- Distributed computing applications - these allow apps to use the computer's resources only when it is idle (for example, an online gaming app).

Device Drivers

Device drivers are software that communicate with the operating system and translate data into a format understood by a hardware peripheral device. Without device drivers, a hardware device would be unable to work with a computer - a message such as 'device not recognized' would appear on the screen. As soon as a device is plugged into a USB port (for example, a memory stick, printer or camera), the operating system looks for the appropriate device driver.

Operating System

Operating System provides both the environment in which applications can be run and a useable interface between humans and computer. An operating system also disguises the complexity of computer software and hardware. Common examples of operating systems include: Microsoft Windows, Apple Mac OS, Google Android and Apple IOS (the latter two being used primarily on tablets and smartphones).

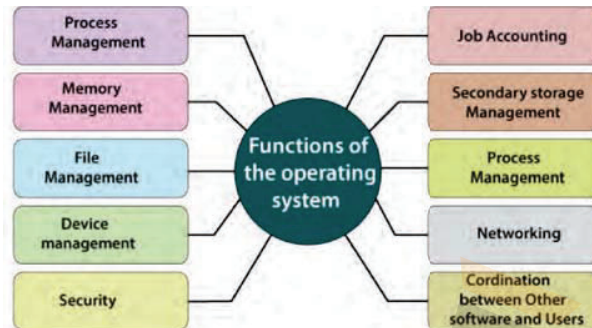


+92-335-8467840



cswithsk@gmail.com

Most computers store the operating system on a hard disk drive (HDD) or solid state drive (SSD) since they tend to be very large programs.



Human Computer Interface (HCI)

Human Computer Interface (HCI) is in the form of a Command Line Interface (CLI) or a Graphical User Interface (GUI).

CLI requires a user to type in instructions in order to choose options from menus, open software, etc. There are often a number of commands that need to be typed in, for example, to save or load a file.

```
~/Work/snyk/snyk$ snyk test

Testing /Work/snyk/snyk...

Organization:   ican
Package manager:  npm
Target file:    package-lock.json
Project name:   snyk
Open source:    no
Project path:   /Work/snyk/snyk
Local Snyk policy:  found
Licenses:       enabled

✓ Tested 455 dependencies for known issues, no vulnerable paths found.
Tip: Detected multiple supported manifests (27), use --all-projects to scan all of them at once.

Next steps:
- Run ' snyk monitor' to be notified about new related vulnerabilities.
- Run ' snyk test' as part of your CI/test.
```

GUI allows the user to interact with a computer (or MP3 player, gaming device, mobile phone, etc.) using pictures or symbols (icons) rather than having to type in a number of commands.



+92-335-8467840



cswithsk@gmail.com

GUIs use various technologies and devices to provide the user interface. One of the most common is WIMP (windows icons menu and pointing device), which was developed for use on personal computers (PC).



Technological Devices increasingly use touch screens and use post-WIMP interactions. With this system, fingers are in contact with the screen allowing actions such as pinching and rotating, which would be difficult to do using a single pointer and a device such as a mouse.

GUI Vs. CLI

Interface	Advantages	Disadvantages
Command Line Interface (CLI)	User is in direct communication with the computer	user needs to learn a number of commands to carry out basic operations
	User is not restricted to a number of pre-determined options	All commands need to be typed in which takes time and can be error prone
	It is possible to alter computer configuration settings	Each command must be typed in using the correct format, spelling, and so on
Graphical User Interface (GUI)	small amount of computer memory	
	user doesn't need to learn any commands	Type of interface uses up considerably more computer memory than a CLI interface
	More user-friendly	User is limited to the icons provided on the screen
	Pointing device (such as a mouse) is used to click on an icon to launch the application this is simpler than typing in commands or a touch screen can be used where applications are chosen by simply touching the icon on the screen	Needs an operating system, such as Windows, to operate, which uses up considerable memory.



+92-335-8467840



cswithsk@gmail.com

Who would use each type of interface?

- **CLI:** a programmer, analyst or technician; basically, somebody who needs to have a direct communication with a computer to develop new software, locate errors and remove them, initiate memory dumps (contents of the computer memory at some moment in time), and so on
- **GUI:** the end-user who doesn't have or doesn't need to have any great knowledge of how the computer works; a person who uses the computer to run software or play games or stores/manipulates photographs, for example.

Memory Management

Memory Management carries out the following functions:

- RAM (Random Access Memory) manages the primary storage and allows data to move between RAM and HDD/SDD during the execution of programs.
- Keeps track of all the memory locations
- Carries out memory protection to ensure that two competing applications cannot use the same memory locations at the same time.

Security Management

It is another part of OS; function of Security Management is to ensure integrity, confidentiality & availability of data. This can be achieved as follows:

- Carrying out operating system updates as and when they become available
- Ensuring that anti-virus software (and other security software) is always up to date, preserving the integrity, security and privacy of data by communicating with.
- Making use of privileges to prevent users entering 'private areas' on a computer that permits multi-user activity this helps to ensure the privacy of data.
- Maintaining access rights for all users
- Offering the ability for the recovery of data when it has been lost or corrupted
- Helping to prevent illegal intrusion into the computer system.

Hardware Peripheral Management

Hardware management involves all input and output peripheral devices.

- Communicates with all input and output devices using device drivers
- Uses a device driver to take data from a file (defined by the operating system) and translates it into a format that the input/output device can understand
- Ensures each hardware resource has a priority so that they can be used and released as required
- Manages input/output devices by controlling queues and buffers.



+92-335-8467840



cswithsk@gmail.com

File Management

- File naming conventions which can be used i.e. filename.docx
- Specific tasks
- Directory structures
- Access Control Mechanisms are maintained
- Memory Allocation

Multitasking

It allows a computer to carry out multiple tasks/processes at a time, sharing resources under the control of OS. For smoothly running all processes OS monitors status of each process.

- Specific time limit
- Interruption
- Priority

Using multitasking management, main memory, HDD/SSD and virtual memory are better managed making the most effective use of CPU time.

User Accounts Management

More than one user can use the Computer System, therefore, each user's data stored in a different part of a memory for security reasons. Operating System is given the task of managing different user accounts.

Application Running

When the computer starts up, part of Operating System needs to be loaded. Start-up of the Computer's motherboard is handled by the basic input/output system (BIOS). BIOS tells computer where the storage devices holds the Operating System can be found; it then loads the part of the Operating System that is needed and executes it.

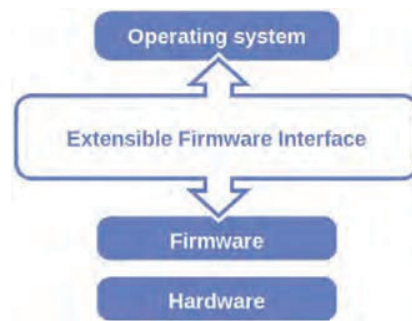
BIOS is referred as firmware. Firmware is a program that provides low level control for devices. BIOS are stored in EEPROM which means its content remain even when the computer is powered off. BIOS can be rewritten, updated or even deleted by a user.



+92-335-8467840



cswithsk@gmail.com

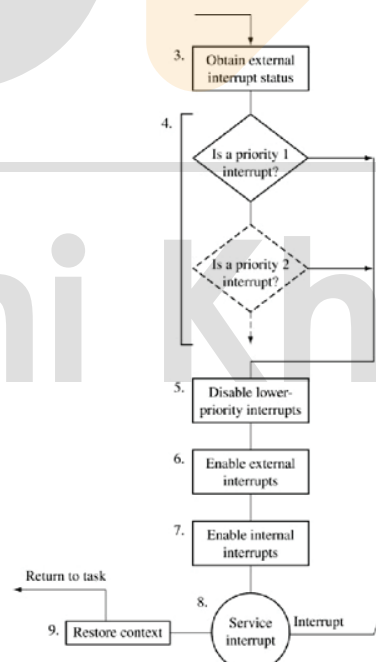


Interrupt

Interrupt is a signal sent from a device or from software to the microprocessor. This will cause the microprocessor to temporarily stop what it is doing so that it can service the interrupt. Interrupts can be caused by:

- Timing signal
- Input/output process
- Hardware fault
- User interaction
- Software errors

Once the interrupt signal is received, the microprocessor either carries on with what it was doing or stops to service the device or program that caused the interrupt. The computer needs to identify the interrupt type and also establish the level of interrupt priority.



Buffers

Temporary storage/memory that holds data being transferred between devices often used to compensate for different speeds of devices examples printer, disk, etc.



Sami Khatri



+92-335-8467840



cswithsk@gmail.com

4.2 Types of Programming Language, Translators & IDEs

Computer only understands machine code, in order to make computer understand the languages programmers, a translator is needed. Through programs we tell computer what to do, when to do and how to do.

Computer program is a set of instructions that enable a computer to perform a specific task. Computer programs can be written in high-level languages and low-level languages depending on the task to be performed and the computer to be used.

High Level Languages

High-level languages enable a programmer to focus on the problem to be solved and require no knowledge of the hardware and instruction set of the computer that will use the program. Many high-level programming languages are portable and can be used on different types of computer.

Low Level Languages

Low-level languages relate to the specific architecture and hardware of a particular type of computer. Low-level languages can refer to machine code, the binary instructions that a computer understands, or assembly language that needs to be translated into machine code.

Sami Khatri



+92-335-8467840



cswithsk@gmail.com

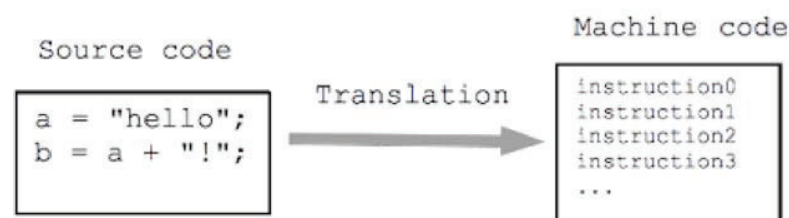
Language	Advantages	Disadvantages
High Level Languages	Independent of the type of computer being used easier to Read, write and understand Programs quicker to write programs. Programs are easier and quicker to debug easier to maintain programs in use	Programs can be larger Programs can take longer to execute Programs may not be able make use of special hardware
Low Level Languages	Special Hardware Special machine-dependent instructions Code that doesn't take up much space in primary memory. Write code that performs a task very quickly.	Longer time to write and debug programs. Programs are more difficult to understand.

Assembly Languages

Assembly language consists of English like symbolic codes known as mnemonics. It is a second generation of language. They represent common strings of machine codes. A language translator, Assembler, is used to convert source code of Assembly language program into object code of machine language. Though Assembly languages are easier than machine language but they are highly detailed and cryptic. So programmers seldom write programs in Assembly language. Instead programmers use Assembly languages to fine-tune important parts of programs written in a high-level language.

Translators

Programs are written by humans in a form that people who are trained as computer programmers can understand. In order to be used by a computer, programs need to be translated into the binary instructions, machine code, that the computer understands. Humans find it very difficult to read binary, but computers can only perform operations written in binary.



+92-335-8467840



cswithsk@gmail.com

A program must be translated into binary before a computer can use it; this is done by a utility program called a translator.

Interpreters

An interpreter is a computer program that reads a statement from a program written in a high-level language, translates it, performs the action specified and then does the same with the next statement and so on.

Assemblers

An assembler is a computer program that translates a program written in an assembly language into machine code so that it can be directly used by a computer to perform a required task. Once a program is assembled the machine code can be used again and again to perform the same task without re-assembly.

Pros & Cons

Translators	Pros	Cons
Interpolator	easier and quicker easier to edit	Cannot run without interpreter
Compiler	Stored ready Without compiler Less space in memory Shorter time	Longer time

Sami Khatri



+92-335-8467840



cswithsk@gmail.com

Integrated Development Environment

An Integrated Development Environment (IDE) is used by programmers to aid the writing and development of programs. There are many different IDEs available; some just support one programming language, others can be used for several different programming languages. You may be using PyCharm (for Python), Visual Studio (for Visual Basic) or BlueJ (for Java) as your IDE.

Features

- Code Editors
- Translator
- Runtime Environment with Debugger
- Error Diagnostics
- Auto Completion
- Auto Correction
- Auto Documenter & Pretty Printing

Code Editors

A code editor allows a program to be written and edited without the need to use a separate text editor. This speeds up the program development process, as editing can be done without changing to a different piece of software each time the program needs correcting or adding to.

Translators

Most IDEs usually provide a translator, this can be a compiler and/or an interpreter, to enable the program to be executed.

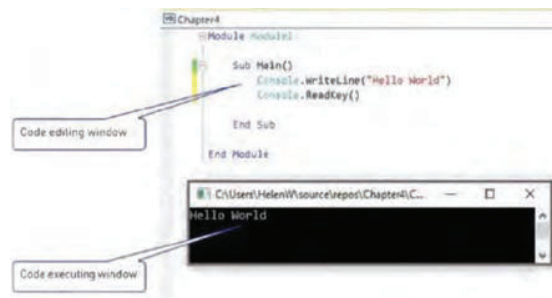
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+92-335-8467840



cswithsk@gmail.com

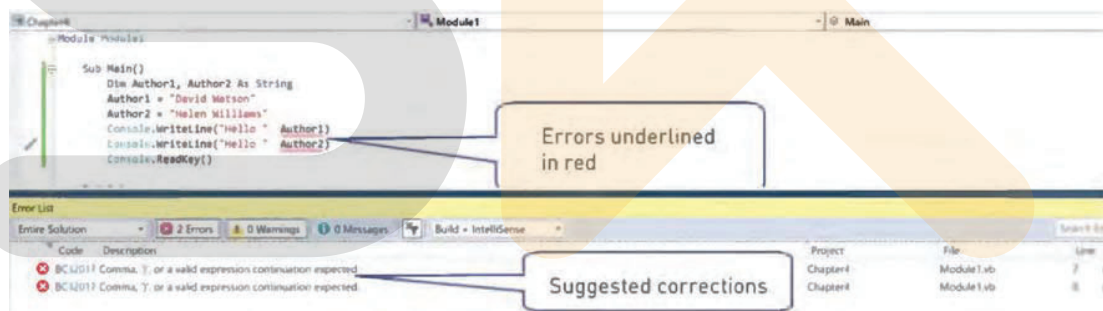


Runtime Environment

A debugger is a program that runs the program under development and allows the programmer to step through the program a line at a time (single stepping) or to set a breakpoint to stop the execution of the program at a certain point in the source code.

Error Diagnostics & Detection

Dynamic error checking finds possible errors as the program code is being typed, alerts the programmer at the time and provides a suggested correction. Many errors can therefore be found and corrected during program writing and editing before the program is run.



Auto Completion

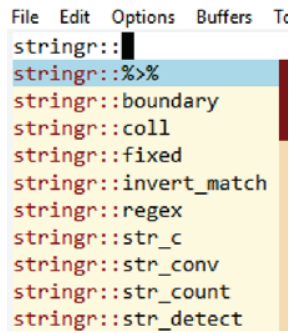
Code editors can offer context-sensitive prompts with text completion for variable names and reserved words.



+92-335-8467840



cswithsk@gmail.com



```
File Edit Options Buffers To
stringr::
stringr::%>%
stringr::boundary
stringr::coll
stringr::fixed
stringr::invert_match
stringr::regex
stringr::str_c
stringr::str_conv
stringr::str_count
stringr::str_detect
```

Auto Documenter & Pretty printing

IDEs can provide an auto-documenter to explain the function and purpose of programming code. Most code editors color code the words in the program and lay out the program in a meaningful way - this is called pretty printing.

Sami Khatri



+92-335-8467840



cswithsk@gmail.com